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REVIEW ARTICLE

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The neurobiology of political ideology: Theories, findings, and future directions

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Abstract

Social scientists are increasingly interested in studying the psychological bases of political preferences. Research at the interface of neuroscience and political psychology is uniquely positioned to test theories that link political attitudes to cognitive, affective, perceptual, and motivational processes in the brain. In this article, we review existing theories and evaluate findings from the growing field of political neuroscience with an emphasis on four brain regions that have emerged as important neural substrates of political ideology: the amygdala, the insular cortex, the anterior cingulate cortex, and the prefrontal cortex. We conclude by discussing current methodological and theoretical limitations before setting out an agenda for future research.

KEYWORDS
ideology, politics, neuroscience, political neuroscience

1 | INTRODUCTION

The past 2 decades have witnessed an explosion of enthusiasm for uncovering the biological bases of human behavior. Once regarded as controversial even among biologists (Segerstråle, 2000), neurobiological approaches have since spread throughout the social sciences and generated a multitude of new subfields (e.g., Hatemi & McDermott, 2011). The interdisciplinary field of political psychology was an early adopter of biologically informative methods (Wahlke & Lodge, 1972), stemming from its history of examining psychological sources of political attitudes (Adorno et al., 1950). Neurobiological approaches intersect with numerous topics of interest in the study of political behavior, including political engagement (e.g., Fowler & Schreiber, 2008), voting (e.g., Rule et al., 2010), and partisan preferences (e.g., Schreiber et al., 2022). Here, we focus on a nascent research program that applies the methods and theories of cognitive and social neuroscience to the study of political ideology.

Although neuroscience methods are relatively new to the study of politics, the idea that human behavior is driven by unconscious mental processes has deep roots in the study of ideology (e.g., Adorno et al., 1950; Fromm, 1941).

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After a period of disinterest, this perspective saw a major revival in both political science and psychology around the turn of the 21st century (Cacioppo & Visser, 2003; Jost et al., 2003), culminating in the direct application of methods from cognitive neuroscience to the study of political ideology (e.g., Amodio et al., 2007). In this review, we build on previous overviews of the political neuroscience literature (e.g., Haas et al., 2020; Jost et al., 2014) while simultaneously narrowing our scope of inquiry to the topic of political ideology (Feldman, 2013).[1] In doing so, we aim to provide an up-to-date overview of research on the neurobiology of ideology.

2 | THE PSYCHOLOGY OF IDEOLOGY

An ideology is a cluster of related beliefs and attitudes about society. For instance, a person who believes that the environment is in danger, cares deeply about this issue, and sees opposing fracking and supporting green energy as the best ways to address it would be said to have an "environmentalist" ideology. In theory, there could be as many ideologies as there are aspects of society about which people disagree. In practice, however, people's political attitudes tend to cluster into one or two ideological dimensions. Some studies find that people who are liberal (or conservative) on one issue tend to be liberal (or conservative) on most other issues (e.g., Azevedo et al., 2019). This implies that most people vary along a single liberal-conservative (or left-right) spectrum that simultaneously contrasts (1) challenging versus maintaining traditional social arrangements and religious morals and (2) rejecting versus accepting inequality (Jost et al., 2009).

Other studies find that people vary along two relatively distinct dimensions of ideology (Malka et al., 2019; Treier & Hillygus, 2009). One of these dimensions, referred to as *economic ideology*, captures people's beliefs about the necessity of combatting socioeconomic inequality. Examples of issues that fall under economic ideology are beliefs about the proper amount of taxation, government efforts to achieve equality of outcomes in domains such as education and health, and funding for social welfare programs such as Medicaid, unemployment insurance, and food stamps. The other dimension, referred to as *social ideology*, captures the extent to which people believe in enforcing traditional social norms and religious morality at the expense of individual autonomy. Examples of issues that fall under social ideology are beliefs about abortion, same-sex marriage, the legality of recreational drugs, and laws regulating acceptable forms of speech and protest.

To measure unidimensional ideology, researchers usually ask respondents to place themselves on scales that range from extremely liberal or left-wing to extremely conservative or right-wing (Jost, 2006). Another measure that appears frequently in the political neuroscience literature is system justification, a psychological orientation to perceive the status quo as good and desirable that is typically associated with conservative ideology (Jost & Hunyady, 2005). When it comes to measuring social and economic ideology, there is less consensus. Political scientists prefer to ask people about their attitudes toward specific social or economic policies (e.g., Feldman & Johnston, 2014). In contrast, psychologists often use the Right-Wing Authoritarianism (RWA) scale to measure social conservatism and the Social Dominance Orientation (SDO) scale to measure economic conservatism. Like system justification, these scales were originally designed to measure psychological precursors to ideology—preferences for social conformity and social hierarchy, respectively. Yet, researchers have found that RWA and SDO are fairly reliable measures of social and economic ideology (Claessens et al., 2023; Forsyth et al., 2023).

Research on the psychological underpinnings of ideology expanded in the decades following World War II, shifting from a psychoanalytic framework to one that emphasized personality and cognition (e.g., Adorno et al., 1950; Allport, 1954; McClosky, 1958; Rokeach, 1960; Sidanius, 1985; Tetlock, 1983; Tomkins, 1963; Wilson, 1973). Consistent with this early research, Jost et al. (2003) theorize that both economic and social conservatism are rooted in psychological needs—*epistemic* needs for order and certainty as well as *existential* needs for safety and security. In this view, individuals who are chronically attentive to threat and averse to uncertainty tend to endorse tradition, religion, and laissez-faire economics because conservative belief systems do a better job than liberal belief systems in explaining and justifying societal hierarchies. These ideological rationalizations of inequality are thought to serve

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a palliative psychological function of reducing uncertainty and anxiety (Goudarzi et al., 2020; Jost & Hunyady, 2005). In this vein, several interrelated findings suggest that conservative ideological orientation is associated with greater psychological management of threat and uncertainty: Amodio et al. (2007) argue that lower sensitivity to cognitive conflict among conservatives leads to a lack of updating in response to information that contradicts existing beliefs; Janoff-Bulman (2009) argues that conservatives exhibit a psychological orientation towards avoiding potential negative outcomes that motivates preservation of the status quo; and Inbar and colleagues (2012) argue that conservatives are more sensitive to disgust, which results in harsh moral judgments against people deemed unusual or abnormal.

Although most research on ideology seeks to explain conservative belief systems, some theories have emerged that seek to explain the motivational bases of liberal ideologies. For instance, McCrae (1996) argues that social liberalism is motivated by a desire to engage with novelty, which motivates experimentation with different ways of arranging society and the pursuit of novel and diverse experiences (see also Carney et al., 2008). Similarly, Janoff-Bulman (2009) theorizes that liberalism is motivated by a psychological predisposition towards approaching rewards that makes the potential benefits of reform more salient than the potential costs. Finally, some recent work argues that left-wing economic ideology is motivated in part by sensitivity to emotional distress that makes harm and injustice feel aversive (Bondü & Inerle, 2020; Panish & Delton, 2023).

While specific theories vary, researchers generally agree that ideology is produced by an interaction between cultural context – a "socially constructed superstructure" – and individual differences in temperament and cognition – a "motivational substructure" (Jost et al., 2009). But because many cognitive and affective processes operate outside of conscious awareness, it is difficult to draw strong conclusions about the psychology of ideology by relying solely on self-reports. Therefore, studies that measure these processes as they manifest in brain structure and function represent a crucial source of information. Evidence from social neuroscience suggests that these individual differences are likely instantiated in the brain (DeYoung et al., 2010; Kanai & Rees, 2011). Thus, the notion that ideology is motivated in part by basic psychological and cognitive processes (Jost et al., 2009) inevitably leads to questions about how neurobiology may influence the development of political attitudes. In the following section, we describe the primary methods used in political neuroscience research to elucidate the neural processes that support ideological outcomes.

3 | NEUROSCIENCE METHODS

Research on the neurobiological basis of ideology has primarily focused on three types of evidence: (1) gray matter volume of specific brain regions; (2) magnitude of activation in specific brain regions in response to stimuli or tasks, as proxied by blood-oxygen levels or electrical activity; and (3) impairment of specific brain regions in lesion patients.

Regional gray matter volume is assessed via structural magnetic resonance imaging (MRI), a tool that can generate detailed two-dimensional images of the tissue structure of individuals' brains (akin to virtual slices). Gray matter volume is a good proxy for neuron count in most parts of the brain (with the exception of the occipital lobe; Ribeiro et al., 2013). Although brain structure is malleable over time, it does not fluctuate significantly moment-to-moment, so it is often considered a useful neurobiological index of relatively stable interindividual differences, such as political orientation (e.g., Kanai et al., 2011).

In contrast, brain *activity* fluctuates constantly, so techniques that assess variability in neural activation are better candidates for capturing how important social and political context can shape brain activity in the moment. One common tool is functional magnetic resonance imaging (fMRI), which uses changes in oxygen levels in blood that are associated with neuronal activation to track activity over time and across regions of the brain. Because fMRI exploits the same detailed imaging capabilities of structural MRI, it boasts high spatial resolution. However, because blood-oxygen levels do not covary instantaneously with neuronal activity, fMRI provides relatively weak temporal resolution (i.e., changes detected on the order of seconds).

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Another common tool for assessing brain activity is electroencephalography (EEG), which makes use of a series of metal nodes placed across the scalp to measure the magnitude and location of the electrical signals conducted across neurons. Unlike fMRI, EEG offers excellent temporal resolution on the order of milliseconds. Using EEG, researchers have identified a number of distinct electrical signals (called event-related potentials, or ERPs) that appear in concert with specific cognitive processes. The downside of EEG relative to fMRI is that its spatial resolution is limited to what can be inferred from the strength of the electrical signals as they reach the various nodes distributed across the scalp. However, studies that observe ERPs in concert with fMRI have helped to pinpoint their neurological substrates (e.g., Liu et al., 2012).

And finally, brain lesion studies examine patients with damage to specific areas of the brain (due to neurological conditions like Urbach-Wiethe syndrome or surgical lesioning as part of medical treatment for conditions like epilepsy). Brain lesion patients are typically compared to matched samples of control populations (without lesions in the same neural region). Such comparisons allow researchers to assess whether a certain brain region is likely necessary for some cognitive function or behavioral outcome, in that individuals with damage in that region exhibit different responses from individuals who do not have damage in that same region (Fellows et al., 2005). Like structural MRI, lesion studies tend to capture relatively more stable interindividual differences in neural anatomy and associated outcomes (e.g., Nam et al., 2021).

4 | THE ROLE OF THE SALIENCE NETWORK AND THE PREFRONTAL CORTEX

Theories about the psychological bases of political ideology implicate a broad range of cognitive and affective processes, including cognitive engagement, updating, approach/avoidance orientation, salience detection, fear, disgust, and anxiety. Although these processes are diverse, in terms of neural structure and function they are far from random. Each of these processes is undergirded, in part, by one of two highly interconnected brain structures—the salience network (SN) and the prefrontal cortex (PFC; Menon, 2015; Uddin et al., 2019). The SN attaches emotional significance to incoming sensory information according to current goals, contexts, and cues. It also engages higher level cognitive processes in the PFC to interpret unexpected events and revise plans in light of new information (Lieberman, 2007; Lieberman et al., 2003). Thus, the SN plays a critical role in determining how we react emotionally to the world around us.

Though the SN is vast and includes many nodes, we suggest that there are three regions (not including the PFC) that are especially relevant for understanding the "motivational substructure" (Jost et al., 2009) of ideology—the amygdala, the insular cortex (insula), and the anterior cingulate cortex (ACC). The amygdala is the most frequently implicated brain region in the political psychology literature (Nam, 2020; Schumacher et al., 2023). This is, in part, because the amygdala serves as a sort of "gateway" to the SN and hence plays a crucial role in all aspects of salience perception. The amygdala's role in motivating hesitation in response to uncertainty and recruiting attention to potential threats *and* rewards makes it a prime candidate for theories that argue that conservative ideology is motivated by aversion to ambiguity, uncertainty, and novelty (e.g., Hibbing et al., 2014; Jost et al., 2003). The insula is important for experiences that draw on internal bodily sensations such as disgust, empathy, and pain (Craig, 2009; Lamm & Singer, 2010; Singer et al., 2009). Disgust sensitivity has been linked across many studies to social conservatism (e.g., Shook et al., 2018) and empathic responsivity to economic liberalism (e.g., Sidanius et al., 2013), which suggests the insula plays an important role in our orientations toward others. The ACC, too, is potentially relevant for both social conservatism and economic liberalism. Not only does the ACC trigger processes of cognitive updating in response to conflicts, but its mechanism for halting behavioral action and engaging the PFC is scaffolded on top of a more basic system for generating emotional distress in response to potential punishment or lack of expected reward (Amodio et al., 2008). As a result, theories that link conservative ideology with resistance to belief updating and those that link liberal economic ideology with sensitivity to emotional distress and anxiety both implicate the ACC (Amodio et al., 2007; Panish & Delton, 2023).

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Beyond the SN, we suggest that a primary candidate for the "motivational substructure" of ideology is the prefrontal cortex. The PFC is integral to processes involving cognition and motivation, such as resolving conflicting goals, searching for objects, exploring environments, and both interpreting and manipulating semantic information. Thus, theories that link liberalism to either a desire to engage with novelty because it is aesthetically pleasing or a tendency to approach stimuli to identify potential innovations both implicate the brain's prefrontal cortex (Janoff-Bulman, 2009; McCrae, 1996).

The various nodes of the SN and the PFC communicate with one another constantly to coordinate the basic actions that we perform every day, like finding our keys and navigating social interactions. One result of this interconnectedness is that it is often not clear why a certain part of the brain "lights up" in a neuroscience study while other closely related regions do not. In many of the studies reviewed in the following sections, a particular task—for instance, exposure to counter-attitudinal information—may trigger activity in, say, the insula, but not the ACC. While some patterns of activation are robust—for instance, the activation of the amygdala in response to threats and the ACC in response to incongruency and conflict—others are less well characterized, and the mapping between tasks and regions is rarely one-to-one. Therefore, we often refrain from commenting on the stimulus used in a particular study, other than to infer that the neural regions of interest attributed salience to the stimulus. We report the results of existing neuroscience studies of ideology in Table 1. As this summary demonstrates, the four brain regions implicated here—the amygdala, the insula, the ACC, and the PFC—are well represented.

5 | THE AMYGDALA: AROUSAL, THREAT, AND UNCERTAINTY

The amygdala is one of the primary subcortical nodes of the SN. It is responsible for assigning salience to and directing attention toward ambiguous, unpredictable, and threatening stimuli, complex social cues such as emotions and social status, and objects that are relevant to an individual's goals in specific contexts (Adolphs, 2010; Gothard, 2020). It also plays a crucial role in motivating hesitation in response to uncertainty (Blanchard et al., 2011; Harrison et al., 2015). In sum, the amygdala keeps us grounded in our environment by orienting us toward objects warranting further inspection and alerting us to their aversive or appetitive status.

Several perspectives on the motivational basis of ideology imply an important role for the amygdala. On the one hand, Jost et al. (2003, 2007, 2012, 2017) argue that people who are sensitive to threat, novelty, and uncertainty—tendencies guided in large part by the amygdala—find the status quo comforting and seek to defend it by advocating for conservative policies. Furthermore, Hibbing et al. (2013, 2014) present evidence that social conservatism reflects a tendency towards preferentially perceiving and responding to negatively valenced stimuli. On the other hand, recent replication attempts find ideological symmetries in psychophysiological manifestations of negativity bias (e.g., Johnston & Madson, 2022). Other psychological theories of ideology take a different tack while still implicating the amygdala's unique properties in political ideology. Tritt et al. (2013, 2014, 2014b, 2016) propose that the adoption of socially conservative ideology is motivated by the heightened perception of stimuli as arousing, leading to a desire for social control of behaviors that evoke strong feelings—regardless of whether their valence is negative (e.g., committing a robbery) or positive (e.g., wearing sexually provocative clothing; see also Leone & Chirumbolo, 2008; Tomkins, 1963). Tritt and colleagues' findings are consistent with Carver's (2005) argument that cautious and avoidant behavior may be a product of perceiving stimuli as highly motivationally salient *in general*, independent of whether those stimuli arouse negative emotion. These findings highlight the need to consider how contextual factors—that is, the features of the environment or the individual that render *either* negative or positive information as particularly salient—may be important in understanding the relationship between motivation, affect, and ideology (Nam, 2020).

Existing research offers strong support for a link between amygdala structure and political ideology. Structural MRI studies assessing gray matter volume find that amygdala size predicts conservative identification (Kanai et al., 2011; Schumacher et al., 2023), RWA and SDO (Balagtas et al., 2023), and system justification (Nam et al., 2018). This latter finding lends support to the idea that heightened attention to uncertainty—mediated by amygdala functioning—encourages people to support existing social arrangements (Nam, 2020).

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TABLE 1 Summary of political neuroscience studies that report a relationship between specific neural regions and political ideology.

Study	Type	Sample characteristics	Tasks/Stimuli/ Conditions	Brain measures	Brain regions [(+) = omitted for space]	Ideology measures	Results ($p < 0.05$; $p < 0.10$)
Amodio et al. (2007)	EEG	US Student (N = 41)	Go/No-Go task	ERN; N2	dACC	Conservative ID	r = −0.59; −0.41
Chiao et al. (2009)	fMRI	US Student (N = 14)	Images of people suffering	BOLD - suffering v not	ACC; L anterior insula	SDO	r = −0.81; −0.80
Inzlicht et al. (2009)	EEG	CA Student (N = 22)	Stroop task	ERN	ACC	Conservative ID	r = −0.01 (non-sig.)
Cheon et al. (2011)	fMRI	KR and US Student (N = 27)	Images of people suffering	BOLD - ingroup v Outgroup	L Temporo-parietal junction	SDO	r = 0.53
Kanai et al. (2011) (1)	MRI	UK Student (N = 90)		Grey matter volume	ACC; R amygdala	Conservative ID	r = −0.27; 0.23
Kanai et al. (2011) (2)	MRI	UK Student (N = 28)		Grey matter volume	ACC; R amygdala; L insula; R Entorhinal	Conservative ID	r = −0.49; 0.55; 0.38; 0.61
Asp et al. (2012)	Lesion	US Clinical (N = 36) and US community (N = 29)		Brain damage type	vmPFC	RWA	Cohen's D = 1.05
Cox et al. (2012)	fMRI	US Student (N = 54)	Resting state	BOLD connectivity	dACC–R Insula/Putamen	System justification	r = −0.33
Graham et al. (2012)	EEG	US Student (N = 65)	Pro/Con moral statements	LPP; N400	Amygdala (inferred from prior research)	Conservative ID	No direct contrast
Lewis et al. (2012)	MRI	UK community (N = 70)		Grey matter volume	Subcallosal gyrus; anterior insula L dmPFC; L Postcentral gyrus; Precuneus	Binding MFs Individualizing MFs	r = 0.57; 0.45 r = 0.47; −0.52; −0.47
Newman-Norlund et al. (2013)	fMRI	US Student (N = 24)	Resting state	BOLD connectivity	L IFG–L SMG IFG–ANG	Party identification	Republican > Democrat Democrat > Republican
Schreiber et al. (2013)	fMRI	US community (N = 82)	Gambling task	BOLD - Winning risky bets	L posterior insula R amygdala	Party identification	Dem > Rep Rep > Dem Total R^2 = 0.53

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TABLE 1 (Continued)

Study	Type	Sample characteristics	Tasks/Stimuli/ Conditions	Brain measures	Brain regions [(+) = omitted for space]	Ideology measures	Results ($p < 0.05$; $p < 0.10$)
Weissflog et al. (2013)	EEG	CA Student (N = 32)	Go/No-Go task	ERN	dACC	RWA; egalitarianism	r = −0.37; 0.40
				N2	vACC	RWA; Egalitar.; Con. ID	r = −0.36; 0.47; −0.45
Ahn et al. (2014)	fMRI	US community (N = 83)	Disgusting Pictures	BOLD activity	R anterior insula; L S2/Post. Insula; (+)	Ideology Composite	Lib > Con
					L amygdala; L dlPFC; (+)		Con > Lib Split sample pred. r = 0.44
Tritt, Inzlicht, & Peterson (2014)	EEG	CA Student (N = 29)	Anticipation-Timing task	FRN - positive feedback	dACC (inferred from prior research)	System justification	$R^2 = 0.09$
Cazzato et al. (2015)	fMRI	IT Student (N = 28)	Gaze-cueing task	BOLD - gaze Incongruity	SMA/Mid-cingulate; anterior insula (+)	SDO	β = −1.4 (left party voters)
Dennis et al. (2015)	EEG	US community (N = 33)	Angry faces × Flanker task	N2 - incongruent Trials [Child]	dACC (inferred from prior research)	Conservative ID [parent]	r = −0.39
Tritt et al. (2016)	EEG	CA Student (N = 42)	Low-arousal pleasant images	LPP		Social conservatism	r = 0.35
Wendell (2016)	EEG	US community (N = 51)	Resting state	Left frontal asymmetry	PFC (inferred from prior research) dACC (inferred from prior research)	Wilson-Patterson scale	r = 0.355
			Go/No-Go task	N2		Social conservatism	r = 0.293
Haas et al. (2017)	fMRI	US community (N = 58)	Candidate policies	BOLD - ingroup Incongruity	ACC/Paracingulate; L Insula/ Lateral OFC	Conservative ID	Liberal > conservative
Ligneul et al. (2017)	fMRI	FR Student (N = 28)	Hierarchy encoding task	BOLD - high v low Rank	R anterior dlPFC	SDO	r = 0.54
Nash et al. (2017)	MRI	CH Student (N = 56)		Grey matter volume	ACC; R dlPFC; R vlPFC; L vlPFC	Binding foundations	r = −0.49; −0.55; −0.61; −0.53
Nam et al. (2018) (1)	MRI	US Student (N = 48)		Grey matter volume	Amygdala	System justification	β = 0.139

(Continues)

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TABLE 1 (Continued)

Study	Type	Sample characteristics	Tasks/Stimuli/ Conditions	Brain measures	Brain regions [(+) = omitted for space]	Ideology measures	Results ($p < 0.05$; $p < 0.10$)
Nam et al. (2018) (2)	MRI	US Student (N = 45)		Grey matter volume	Amygdala; ACC; L insula	System justification	β = 0.327; −0.232; *0.122*
Pedersen et al. (2018)	fMRI	US Student (N = 24)	Threat of electric shock	BOLD connectivity	L Amygdala–BNST	Economic conservatism	β = 0.556
			No threat				β = −0.511
Kim et al. (2020)	fMRI	KR Student (N = 102)	Resting state	BOLD connectivity	ACC–L SFG/dlPFC; L Insula–L OCP; (+)	Conservative ID	Conservative > liberal
	MRI			Grey matter volume	R SOG; R Precentral gyrus		Conservative > liberal
Krosch et al. (2021)	fMRI	US Student (N = 41)	Mixed-Race faces	BOLD - racial ambiguity	Anterior insula	Conservative ID	r = 0.42
Nam et al. (2021)	Lesion	US Clinical (N = 62)		Brain damage type	Frontal lobe dlPFC	Conservative ID	b = 1.9 (Δ Adj. R^2 = 0.09)
		- Subsample (N = 43)		Brain damage magnitude			b = 0.15
Nash & Leota, (2022) (1)	EEG	NZ Student (N = 60)	Auditory Startle × threat	P3		Conservative ID	Con = Lib (non-sig.)
Nash & Leota, (2022) (2)	EEG	CA Student (N = 100)	Auditory Startle × threat	P3	dACC; L dlPFC; Parietal cortex; (+)	Conservative ID	Conservative > liberal
Yang et al. (2022)	fMRI	US community (N = 174)	Reward	BOLD connectivity	Amygdala; R ACC; (+)	Conservative ID	Conservative > liberal
			Retrieval; empathy		R amygdala; (+)		Conservative > liberal
Yun et al. (2022)	EEG	KR Student (N = 17)	Ethical Transgression	N100; N400	Middle frontal gyrus	Conservative ID	Liberal > conservative
				LPP	Precuneus		Conservative > liberal

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TABLE 1 (Continued)

Study	Type	Sample characteristics	Tasks/Stimuli/ Conditions	Brain measures	Brain regions [(+) = omitted for space]	Ideology measures	Results ($p < 0.05$; $p < 0.10$)
Balagtas et al. (2023)	MRI	SG Student (N = 82)		Grey matter volume	Amygdala vmPFC	RWA; SDO	+RWA; +SDO −RWA; −SDO
					Anterior insula		−SDO
Schumacher et al. (2023)	MRI	NL population (N = 928)		Grey matter volume	Amygdala; Fusiform gyrus	Conservative ID	r = 0.068; 0.081

Note: The italic values are significant at the $p < 0.10$ level.

Study Types: EEG, Electroencephalography; fMRI, Functional Magnetic Resonance Imaging; MRI, Structural Magnetic Resonance Imaging.

Sample Countries: CA, Canada; CH, Switzerland; FR, France; IT, Italy; KR, Republic of Korea; NL, Netherlands; NZ, New Zealand; SG, Singapore; UK, United Kingdom; US, United States.

Measures: BOLD, Blood-Oxygen-Level-Dependent Imaging; ERN, Error-Related Negativity; LPP, Late Positive Potential; RWA,Right Wing Authoritarianism; SDO, Social Dominance Orientation.

Brain Region: ACC, Anterior Cingulate Cortex; ANG, Angular gyrus; BNST, Bed Nucleus of the Stria Terminalis; dACC, Dorsal ACC; dlPFC, Dorsolateral PFC; dmPFC, Dorsomedial PFC; Inferior frontal gyrus = IFG; OCP, Occipital Pole; PFC, Prefrontal Cortex; SFG, Superior Frontal Gyrus; SMG, Supramarginal gyrus; SOG, Superior occipital gyrus; S2, Secondary Somatosensory Cortex; vACC, Ventral ACC; vlPFC, Ventrolateral PFC; vmPFC, Ventromedial PFC.

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Brain imaging studies have compared how liberals and conservatives respond to stimuli that are theorized to engage the amygdala. Using fMRI to assess regional brain activity, researchers have found that Republicans and conservatives exhibit greater activation of the amygdala in response to risky bets (Schreiber et al., 2013) and disgusting images (Ahn et al., 2014), respectively. Another fMRI study by Pedersen et al. (2018) finds that economic conservatives display greater connectivity between the amygdala and the bed nucleus of the stria terminalis—a key region for motivating defensive responses to uncertain threats (Klumpers et al., 2017)—in response to the threat of electrical shocks.[2] Similarly, an fMRI analysis of functional connectivity (i.e., the co-activation of specific subregions of the brain) finds that conservatives exhibit more amygdala activity than liberals during tasks involving reward, empathy, and working memory (Yang et al., 2022). These studies suggest that a wide variety of stimuli evoke greater amygdala activity in conservatives, potentially consistent with the view that conservatism is associated with heightened perception of motivational salience rather than negative valence per se (Tritt et al., 2013, 2014a, 2014b).

EEG studies provide indirect evidence for the amygdala-ideology link by measuring the amplitude of the Late Positive Potential (LPP)—an ERP that is associated with orienting attention towards salient objects and that often (but not always) co-occurs with activity in the amygdala and insula (Liu et al., 2012). Consistent with structural MRI and fMRI studies, Yun et al. (2022) find that conservative identification predicts larger LPPs when faced with a hypothetical moral quandary. In contrast, however, Graham et al. (2012) find that liberals and conservatives both exhibit LPPs in response to counter-attitudinal moral statements. Because individuals might perceive the stimuli used in these studies as simultaneously threatening, negative, and salient, these results are broadly supportive of the theories laid out above but do not allow us to adjudicate among them. Two studies that attempt to do so by varying the content of their stimuli yield conflicting results. An EEG study by Tritt et al. (2016) finds that social conservatism predicts larger LPPs in response to low-salience pleasant stimuli, but not to aversive stimuli or high-salience pleasant stimuli. The authors interpret these results as indicating that conservatives have a lower threshold for perceiving stimuli as salient, but do not react more strongly to high-salience stimuli. Yet, Ahn et al. (2014) find the opposite: conservatives show greater activity in the amygdala in response to high-salience disgusting images, but not low-salience pleasant images.[3]

As the contrasting results of these four studies demonstrate, we currently know little about how or why the motivational salience encoded by the amygdala motivates conservative ideology. Future research will need to use creative experiments to investigate which aspects of salience are mainly responsible for mediating the relationship between the amygdala and conservative ideology. This will need to involve separating the effects of salience, negative valence, social threat, uncertainty, and ambiguity—a challenging task that, if successful, will massively increase our understanding political ideology.

6 | THE INSULAR CORTEX: DISGUST, EMPATHY, AND PAIN

The insula plays an important role in integrating and interpreting various dimensions of emotional experience. It receives audio-visual inputs from the occipital and temporal lobes, saliency inputs from subcortical regions including the amygdala, and signals of autonomic processes and bodily sensations such as temperature and pain from the body's nerve endings (Craig, 2009). Thanks to this unique combination of external and internal inputs, the insula can convey emotional experiences that blend interoceptive awareness of one's own body and perceptions of external objects (Lamm & Singer, 2010). Building on this faculty, the insula forms predictive models of affect in relation to body states and uses these to both anticipate the emotional consequences of actions and to infer what other people's internal states feel like based on their behavior (Craig, 2009; Singer et al., 2009). Two emotion states that draw heavily on this unique property of the insula are empathy, in which the sight of another person's affectively-charged experience triggers a reminder of the associated body state (Jabbi et al., 2007), and core disgust, in which images or scenarios that imply the potential for biological contamination—including the disgust responses of other people—trigger a body state that motivates avoidance and purging (Chapman & Anderson, 2012; Jabbi et al., 2008).

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Research from social and evolutionary psychology suggests that disgust helps to shape political attitudes in the social domain (see Terrizzi et al., 2013, for a review). According to this theory, the human mind has evolved to interpret non-normative behaviors and appearances as heuristic cues of infection risk, leading individuals who are dispositionally disease-averse or who have been exposed to infection cues to develop prejudiced attitudes toward those perceived as physically or behaviorally unusual (Shook et al., 2018). Recent research in political psychology suggests there is a link between exposure and aversion to infectious diseases and conservative attitudes and vote choice (Nam et al., 2022; O'Shea et al., 2022; Zmigrod et al., 2021). However, some research also indicates that disgust is related to left-wing economic ideology (Kam & Estes, 2016; Petrescu & Parkinson, 2014). This finding may be due to the insula's involvement in both disgust and, via pain perception, empathy (Bernhardt & Singer, 2012; Singer et al., 2004), such that increased insula activity corresponds to greater egalitarianism (Dawes et al., 2012).

In structural MRI studies, insula volume predicts conservative identification (Kanai et al., 2011), system justification (Nam et al., 2018), lower SDO (Balagtas et al., 2023), and endorsement of purity (Lewis et al., 2012), a disgust-related moral foundation that is highly correlated with social conservatism (Hatemi et al., 2019). However, other studies have failed to find a relationship between insula volume and ideology (Kim et al., 2020; Nash et al., 2017; Schumacher et al., 2023). Two fMRI studies corroborate Balagtas et al.'s finding of a negative relationship between the insula and SDO, showing that individuals high in SDO exhibit less insula activity in response to contradictory gaze cues (Cazzato et al., 2015) and images of people experiencing emotional suffering (Chiao et al., 2009). In other fMRI studies, liberalism is associated with heightened insula activity in response to disgusting images (Ahn et al., 2014) and policy incongruency from ingroup political candidates (Haas et al., 2017), while Democrats show greater activity in the posterior insula in response to risky bets (Schreiber et al., 2013). However, Krosch et al. (2021) find that conservatives exhibit more anterior insula activity in response to racially ambiguous faces.

In light of these results, we propose that the insula's interoceptive properties may simultaneously facilitate social conservatism (via disgust) and economic liberalism (via empathy, perhaps bounded by group attachments). Though speculative, this notion is supported by evidence from survey-based research that suggests the potency of social emotions is related to *increased* economic conservatism but *decreased* social conservatism (i.e., libertarianism; Bardeen & Michel, 2019; Iyer et al., 2012).

7 | THE ANTERIOR CINGULATE CORTEX: CONFLICT MONITORING AND EMOTIONAL DISTRESS

Along with the anterior insula, the dorsal ACC (dACC) constitutes the central hub of the salience network (Menon, 2015). Unlike the insula and amygdala, the ACC projects to the motor cortex and spinal cord, giving it direct control over behavior (Asemi et al., 2015; Devinsky et al., 1995). It also boasts extensive reciprocal connectivity with the lateral PFC (lPFC), where plans for how to act in order to achieve goals are devised and stored (Tanji & Hoshi, 2008; Yim et al., 2019). The ACC's unique position at the intersection of the brain's emotional, cognitive, and motor networks prepares it to perform its main function: conflict-monitoring that serves updating and implementation of goal-oriented action plans (Paus, 2001).

Researchers have discovered that these two processes—implementation and updating—generate their own sets of ERPs. An ERP called the N200 (or N2) emerges when the ACC successfully inhibits an action that would have conflicted with a plan (Patel & Azzam, 2005). Another ERP called error-related negativity (ERN) appears when the ACC recognizes that an action has deviated from the lPFC's template for how goal-pursuit is supposed to unfold—in short, when a person's brain realizes that they have made a mistake (Holroyd & Coles, 2002). Importantly, the ERN is believed to correspond to an affective pain signal that is central to the experience of negative emotion, including anxiety, grief, and feelings of distress when other people are harmed (Perkins et al., 2013; Saunders & Inzlicht, 2020; Singer et al., 2004).

The ACC's roles in (1) updating cognitive representations and (2) generating affective pain make it likely that this region is related to political ideology. A longstanding theory holds that conservative ideology is a product of

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resistance to belief-updating, which is said to encourage absolutist judgements and strict adherence to received wisdom (Adorno et al., 1950; Jost et al., 2003; Rokeach, 1960). In accordance with this perspective, social conservatism predicts poorer performance on behavioral measures of cognitive updating (Costello et al., 2023), whereas conservatives outperform liberals in tasks requiring perseverance and consistency (Buechner et al., 2021). Meanwhile, there is growing evidence that the concern with suffering and injustice that characterizes economic liberalism is motivated, in part, by sensitivity to anxiety that increases anticipation of and aversion to harm (Bondü & Inerle, 2020; Helminen et al., 2022; Panish & Delton, 2023).

The political neuroscience literature offers support for a link between the ACC and liberal ideology. In MRI studies, greater ACC volume predicts liberal identification (Kanai et al., 2011), lower system justification (Nam et al., 2018), and diminished endorsement of binding moral foundations (Nash et al., 2017)—a close proxy of social conservatism (Kugler et al., 2014). However, other studies have failed to find a relationship between ACC structure and ideology (Kim et al., 2020; Lewis et al., 2012; Schumacher et al., 2023).

Studies of neural function provide further nuance in the links between the ACC and ideology. In fMRI studies, individuals scoring higher in SDO show diminished ACC activity in response to images of people suffering (Chiao et al., 2009) and diminished mid-cingulate activity during a gaze-cueing task (Cazzato et al., 2015). Meanwhile, liberals show increased ACC activity in response to viewing candidates with counter-stereotypical policy positions (Haas et al., 2017). Similarly, in a series of EEG studies, the magnitudes of ERN and N2 amplitudes generated by the ACC (as well as better performance on a cognitive conflict task) predict liberalism (Amodio et al., 2007), as well as lower RWA, and greater egalitarianism (Weissflog et al., 2013). In a third EEG study, Dennis et al. (2015) find that the magnitude of children's N2 amplitudes during a cognitive conflict task predicts their parent's self-reported liberalism. However, one study finds no relationship between N2 amplitude and ideology (Inzlicht et al., 2009), and another finds that N2 amplitude is correlated with social conservatism (Wendell, 2016).

Finally, fMRI studies of functional connectivity find that connectivity between the ACC and dlPFC predicts conservatism (Kim et al., 2020), whereas connectivity between the ACC and the insula predicts liberalism (Cox et al., 2012). According to Rueter et al. (2018), functional connectivity between the ACC and the dlPFC facilitates higher levels of the personality trait of conscientiousness—characterized by orderliness and industriousness—by increasing the speed and accuracy with which the brain categorizes stimuli as relevant or irrelevant to current goals. Thus, Kim et al.'s results may point to a neural basis for the widely replicated link between conscientiousness and conservative ideology (Sibley et al., 2012). Meanwhile, Cox et al.'s results may point to a neural basis for the finding that liberals show greater interoceptive sensitivity than conservatives (Ruisch et al., 2022).

8 | THE PREFRONTAL CORTEX: APPROACH, AVOIDANCE, AND COGNITIVE EXPLORATION

Whereas the ACC helps to adjust behavior to accomplish goals, the prefrontal cortex (PFC) is where plans for how to pursue goals are devised and refined (Tanji & Hoshi, 2001). The PFC receives inputs from nearly every part of the brain, allowing it to keep track of how behavioral plans are unfolding, integrate relevant information into mental representations, and tinker with these representations to devise and execute new plans and revise old ones that have gone awry (Gehring & Knight, 2000; Yim et al., 2019). Projections to the motor cortex also allow the PFC to take direct control of behavior when tasks require continual thought (Tanji & Hoshi, 2008). In short, the PFC is crucial for both cognitive engagement with ideas and behavioral engagement with objects in one's environment.

8.1 | Cognitive exploration

Research suggests that individual differences in PFC structure and function correspond to differences in both the motivation and the ability to engage in exploratory cognitive activity. The volume of the lPFC and its connectivity with

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the rest of the brain are related to individual differences in both cognitive ability (Cole et al., 2012; Cox et al., 2019) and openness to experience—a personality trait characterized by enthusiastic engagement with new information (DeYoung, 2015; DeYoung et al., 2005, 2010).

Psychologists have long argued that the tendency to engage in cognitive exploration is linked to political liberalism (Adorno et al., 1950). Recent research lends support to this view, with some caveats. Social conservatism is related to lower openness to experience (Osborne et al., 2021), less creativity (Dollinger, 2007), greater cognitive rigidity (Jost, Sterling, et al., 2017), less cognitive reflection (Deppe et al., 2015; Yilmaz & Saribay, 2016, 2017, 2018), lower cognitive ability (Ludeke et al., 2017; Onraet et al., 2015), and less motivation to exert effort on cognitive tasks (Burger et al., 2020; Ludeke et al., 2016). Though these effects are often small, they nonetheless suggest that social conservatism is tied to a distinctly non-exploratory or "closed" cognitive style (Rokeach, 1960).[4] Researchers have proposed a number of reasons why this might be. Frequent reappraisal and deliberation may raise doubts about the validity of attitudes that are based in rapid emotional responses (Eidelman et al., 2012; Feinberg et al., 2014; Lee et al., 2013; Talhelm et al., 2015). People who enjoy cognitive exploration may gravitate towards critical or counter-intuitive ideas that contradict received wisdom, whereas those who find cognitive exploration unpleasant or difficult may avoid such conflict-inducing ideas (McCrae, 1996; Onraet et al., 2011).

Lesion studies that investigate the relationship between the integrity of specific brain regions and ideology have allowed researchers to directly test whether PFC functionality contributes to political ideology. Asp et al. (2012) compare patients with bilateral damage to the ventromedial PFC to patients with non-prefrontal brain damage, non-brain-damaged patient controls, and a non-patient population sample. They find that patients with prefrontal brain damage report substantially higher right-wing authoritarian views than all three of the control groups. Similarly, Nam et al. (2021) find that patients with prefrontal brain damage report greater conservatism than those with non-prefrontal brain damage and non-brain-damaged controls. Using MRI, the authors are able to quantify the extent of lesioning to the dorsolateral PFC among members of the prefrontal brain damage group and demonstrate that increased damage in this region is linearly related to greater conservative identification. These findings are consistent with a structural MRI study by Nash et al. (2017), which finds that endorsement of binding moral foundations is associated with substantially less gray matter in the lateral PFC. Moreover, Lewis et al. (2012) report that larger left dorsomedial PFC volume is related to stronger endorsement of the individualizing moral foundations (harm and fairness) that liberals tend to value more highly, but is unrelated to endorsement of the binding moral foundations (authority, ingroup loyalty, and purity) that conservatives tend to value more highly.

8.2 | Approach/avoidance orientation

The PFC is also related to individual differences in avoidant behavior. In most (right-handed) people, the left PFC specializes in coordinating cognitive and behavioral elaboration, approach, and exploration, while the right PFC mainly serves to inhibit left-hemispheric activity and orient the individual towards salient social cues, ambiguity, and threat (Brancucci et al., 2009; Braun, 2007; Devinsky, 2009; Marinsek et al., 2014). In EEG studies, greater left hemispheric PFC activity corresponds to a tendency to experience approach-oriented emotions, whereas greater right hemispheric PFC activity corresponds to avoidance-oriented emotions (Coan & Allen, 2004; Harmon-Jones, 2003).

Behavioral approach versus avoidance orientation is central to a number of psychological accounts of ideology (e.g., Hibbing et al., 2014; Janoff-Bulman, 2009; Mendez, 2017; Shook & Fazio, 2009). These theories posit that avoidance orientations function to prevent negatively perceived outcomes, leading people to adopt socially conservative ideologies that emphasize control and restraint, whereas approach orientations function to achieve positively perceived outcomes, leading people to adopt liberal ideologies that emphasize reform and experimentation. Thus, there is a strong theoretical expectation that conservatism should correlate with neural signatures of avoidance orientation. Nonetheless, at least one theoretical account predicts the opposite relationship on the basis that conservatism is associated with approach-oriented emotions (such as anger) whereas liberalism is associated

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with avoidant emotions (such as sadness; Vigil, 2010; Vigil & Strenth, 2014; see Feldman & Huddy, 2014; Inbar & Pizarro, 2014; Lilienfeld & Latzman, 2014 for discussions).

Just one study explicitly tests the hypothesis that ideology is related to neural signatures of approach-avoidance orientation. Wendell (2016) reports that left hemispheric asymmetries in resting-state prefrontal activity significantly predict social conservatism—the opposite of what theories linking conservatism to avoidance would predict (cf. Mendez, 2017). Similarly, two neuroimaging studies find a positive relationship between conservative identification and left—but not right—dorsolateral prefrontal activity elicited by disgust (Ahn et al., 2014) and threat cues (Nash & Leota, 2022). Finally, a study using non-invasive brain stimulation of the (bilateral) dlPFC shows that temporarily enhanced dlPFC activity is associated with greater conservatism (Chawke & Kanai, 2016), further suggesting that the relationship between PFC function and ideology is likely dependent on contextual factors, such as an individual's current goals.

9 | RECOMMENDATIONS AND FUTURE DIRECTIONS

The biological pathways identified in political neuroscience can help to clarify theories of ideology and suggest new ways forward for political psychology research. To facilitate continued progress in political neuroscience, we evaluate the strength of the existing evidence, offer methodological considerations that we believe are useful for future studies, and outline some potential future directions for research.

9.1 | State of the field and current theories

Political neuroscience research is expensive and time consuming—as a result, most empirical trends noted in this review are limited to a handful of relatively small studies. Additionally, existing studies tend to under-sample conservatives—particularly extreme conservatives. However, the field has recently taken a major step towards overcoming these limitations. Schumacher et al. (2023) report a pre-registered replication of Kanai et al. (2011) in a sample of 928 Dutch individuals. This is not only the largest sample analyzed in the political neuroscience literature (see Table 1), but also the only one that is representative of a national population in terms of educational attainment and socioeconomic status. These authors successfully replicate Kanai et al.'s (2011) finding that amygdala volume predicts self-reported conservatism, but do not find relationships with insula, ACC, or PFC volume. Another recent pre-registered study conceptually replicates Kanai et al.'s (2011) findings in the political context of Singapore using the RWA and SDO scales (Balagtas et al., 2023). Given these results, we posit that amygdala volume is a robust neurobiological correlate of conservative ideology. This, in turn, should encourage researchers to prioritize studies that can reveal what aspects of amygdala function mediate this relationship. One promising avenue looks at whether aversion toward uncertainty—mediated by the amygdala—causes people to oppose changes to the current social and political system, which in turn leads people to adopt conservative ideologies (Nam, 2020; Nam et al., 2018).

A few other empirical trends, though less well-established, warrant further study. The first of these is the relationship between left-wing ideology and both ACC and insula activity. On the one hand, a series of experimental EEG studies lend support to the idea that ACC functioning promotes liberalism by helping people to update their beliefs in response to cognitive conflict (Amodio et al., 2007; Dennis et al., 2015; Weissflog et al., 2013). However, these results have not always held up (Inzlicht et al., 2009; Wendell, 2016). On the other hand, by virtue of their shared role in conveying emotional pain (Singer et al., 2004), the insula and ACC are both implicated in theories that link anxiety to left-wing economic ideology (Panish & Delton, 2023). In addition to the EEG studies cited above, fMRI studies have repeatedly linked activity in these regions to liberalism (Ahn et al., 2014; Cox et al., 2012; Haas et al., 2017; Schreiber et al., 2013) and to lower SDO in particular (Cazzato et al., 2015; Chiao et al., 2009). However, because only one of these studies was designed to test this hypothesis, the theory should be regarded as speculative. Future work in this

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area should focus on conducting large-scale, pre-registered replications of the experiments conducted by Amodio et al. (2007) and Chiao et al. (2009) to further probe ideological links to the ACC in a variety of contexts.

Another avenue for future research is the relationship between insula volume and social conservatism. Because of its role in conveying disgust (Chapman & Anderson, 2012), there are strong theoretical expectations that insula structure or functioning should underly hostility toward non-normative behaviors (Shook et al., 2018). Some studies do find a link between insula volume and various measures of conservatism (Kanai et al., 2011; Lewis et al., 2012; Nam et al., 2018). However, other studies fail to corroborate this trend (Balagtas et al., 2023; Kim et al., 2020; Nash et al., 2017; Schumacher et al., 2023). Additionally, one study finds that conservatives exhibit greater insula activity in response to racially ambiguous faces (Krosch et al., 2021)—consistent with the heightened aversion toward non-normative appearances predicted by accounts that link conservatism to disgust sensitivity (Shook et al., 2018). Given that the insula may simultaneously facilitate right-wing and left-wing attitudes in different ideological dimensions, existing studies that use unidimensional measures of ideology may underestimate the true relationship between insula volume and social conservatism.

Finally, the relationship between PFC functioning and social conservatism also deserves further study. Though only two studies have tested this theory directly, both have found evidence that PFC damage is related to conservatism (Asp et al., 2012; Nam et al., 2021). Specifically, Asp et al. find a relationship with Right-Wing Authoritarianism and Nam et al. find a relationship with conservative identification. Given that both measures are more closely related to social conservatism than economic conservatism, these results are consistent with the finding that social conservatism is related to a closed cognitive style (Feldman & Johnston, 2014). Given the role of the PFC in cognitive style and motivation, future research may benefit from exploring whether differences in openness to experience and need for cognition mediate the relationship between ideology and brain structure and function.

9.2 | Conceptualization and measurement of ideology

Many studies rely on subjects' self-placement on a liberal-conservative or left-right scale to measure ideology. This approach can be prone to measurement error, in part because the interpretation of concepts like "liberal versus conservative" and "left versus right"—and even extremity—can vary as a function of education and cultural context (Arikan & Ben-Nun Bloom, 2015; Bauer et al., 2017; Ellis & Stimson, 2012). This is not to say that ideological identification is uninformative. Existing work shows that conservative identity in the US corresponds to a unique ideological profile—one that blends resentment against cultural change with some openness to economic intervention (Claassen et al., 2015; Ellis & Stimson, 2012). Researchers would benefit from grounding their use of liberal-conservative identification in this work. At the same time, the high correlation between social and economic conservatism in the United States and United Kingdom means that it is reasonable to use these scales in cases where researchers are working with pre-existing data or are subject to space constraints on their questionnaires (Azevedo et al., 2019).

A broader issue in political neuroscience is the question of whether results obtained with different ideology scales are comparable. When it comes to comparing outcomes from studies that use measures of unidimensional ideology such as liberal-conservative identification and system justification, there is a need for caution. As stated above, it is important to ground interpretations of liberal-conservative identification in work that analyzes the meanings of these labels in particular cultural contexts. It is often a safe assumption that people who self-identify as conservative or right-wing hold socially conservative attitudes on issues like abortion, same-sex marriage, and women's rights. However, the relationship between these labels and economic attitudes is far more variable (Arikan & Ben-Nun Bloom, 2015; Ellis & Stimson, 2012). System justification is even more variable. In the US, it is consistently correlated with conservatism (Jost, Langer, et al., 2017). But in France, for instance, it is related to liberal ideology (Langer et al., 2020). As with liberal-conservative identification, attention to the attitudinal correlates of these scales in particular cultural contexts is necessary. Yet, once this context is taken into account, the distinction between system justification and ideology can help to identify whether opposition to system change mediates relationships between neurobiology and ideology.

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When it comes to studies that measure social and economic ideology, measurement may be more straightforward. There are strong theoretical and empirical foundations for the view that people's ideological convictions can be captured using two dimensions, one concerned with group cohesion and conformity and one concerned with competition and hierarchy (Ashton et al., 2005; Claessens et al., 2020; Feldman & Johnston, 2014). Moreover, large cross-national studies have found that scales that ask people specific social and economic policy questions and the RWA and SDO scales each tap the same underlying preferences for conformity versus autonomy and cooperation versus competition, respectively (Claessens et al., 2023; Forsyth et al., 2023). The same appears to be true for the binding and individualizing moral foundations scales, which are statistically indistinguishable from RWA and SDO, respectively (Kugler et al., 2014). Thus, researchers are on firm ground when comparing outcomes from political neuroscience studies that use either policy attitudes or psychological scales to measure social and economic ideology.

The importance of using fine-grained measures of ideology goes beyond potential differences in how people interpret political labels. Research finds that economic and social ideology have both common and unique dispositional correlates (Ashton et al., 2005; Feldman & Johnston, 2014). In at least some cases, variables that positively predict social conservatism negatively predict economic conservatism (Costello & Lilienfeld, 2021). By asking subjects whether they endorse specific issue positions, researchers can confront these potential sources of ambiguity. It remains unclear in the extant literature the precise psychological, neurobiological, and contextual factors that differentiate social ideology from economic ideology, and future research would do well to interrogate these relationships. In particular, future research should aim to explain why perceiving negative stimuli as more salient is sometimes associated with social conservatism (e.g., Shook & Fazio, 2009) and other times economic liberalism (e.g., Helminen et al., 2022; Panish & Delton, 2023). This could be done, for example, by showing people a series of equally negatively-valenced images while varying the specific content (e.g., poverty, military threats, natural disasters, etc.) and observing whether distinct domains of negative stimuli differentially affect the neural activity of social conservatives and economic liberals (Hatemi & McDermott, 2020).

More broadly, a growing literature has highlighted the important roles of political awareness, cultural norms, and economic development in linking predispositions to ideologies (for a review see Federico & Malka, 2018). Future work in political neuroscience would do well to interrogate these perspectives more directly, which suggest that the motivational theories of ideology that have directed much political neuroscience research may not hold to the same extent, or even in the same direction, for individuals embedded in different ecologies (e.g., Cheon et al., 2011). Although the constraints of the lab setting inherent to neuroimaging techniques put limitations on the realism of the social and political contexts that participants can experience during a study, we suggest two approaches that may help to align ecological validity with the neuroimaging lab context. First, experimental approaches that leverage fMRI or EEG can use real-world political stimuli (including photos and videos) to understand how the brain dynamically responds to salient ideological contexts. Second, we urge researchers to collaborate across geopolitical contexts to clarify whether patterns of neural structure may be consistent or may vary according to the cultural context and its attendant dominant ideologies.

9.3 | Mapping out a future beyond brain mapping

Political neuroscience studies are ideally situated to test theories of political ideology that implicate emotional, cognitive, or perceptual pathways. The findings reviewed here demonstrate the utility of this approach and suggest promising avenues for future research. Existing work has largely pursued a brain mapping approach that seeks to link specific regions of the brain to associated political outcomes. We argue that future work should build on brain mapping studies to use experimental designs to assess the causality of relationships between brain and behavior (see Jost et al., 2014). These experiments can be tailored to known functions of the SN and prefrontal regions that are emerging as core areas of political attitudes, as we have outlined here. For instance, studies focusing on the amygdala could vary experimental treatment by intensity, valence, and content to investigate whether salience, negativity, or

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threat is mainly responsible for mediating the amygdala's relationship with conservative ideology. Future research focusing on the insula may experimentally manipulate both empathy and disgust to check for interdependence or spillover among emotions linked to interoception and their consequences for social versus economic dimensions of ideology. In addition, future work focusing on the ACC, as well as the PFC, should help to clarify conflicting findings linking the structure and function of each of these regions to liberalism in some studies and conservatism in others. We suggest that such clarification can be accomplished in at least two ways.

First, researchers can measure these regions' primary psychological correlates—such as conflict monitoring, cognitive updating, openness to experience, and behavioral activation and inhibition—to test whether their relationships with ideology are in fact mediated by basic indicators of cognitive flexibility. Second, the variability of existing evidence on the relationship between the ACC and PFC and ideology suggests that important contextual, cultural, and motivational factors may moderate the ways in which ACC and PFC function promotes conservatism or liberalism. By measuring or experimentally manipulating these factors, researchers can improve our understanding of how situational and dispositional influences intersect to shape political ideology. As this example illustrates, we believe that neuroscientific studies of psychological mechanism and behavioral studies of political context can achieve consilience and deeper insights regarding political attitudes by working iteratively through a process of "collaborative cross-examination" (Jost et al., 2014). That is, findings from neuroscientific studies (even those that do not directly examine political outcomes) can inform subsequent behavioral studies that test the psychological process within political contexts—*and* findings from such behavioral studies can then inform the next round of neuroscientific investigation. This does not mean that the same research design must be tested with every available neuroscience technique. The idea is that both levels of analysis are necessary to interrogate and expand upon the claims from one another, such that the research occurring at both levels of analysis intersect rather than run in parallel. After all, it is of paramount importance to connect the basic neural processes elucidated in neuroscientific approaches to the consequential political behaviors that affect our societies. In summary, a key goal for political neuroscience should be to refine and test existing theories that link neurobiology to ideology at the nexus of the "motivational substructure" of the brain and the "socially constructed superstructure" of the cultural context (Jost et al., 2009)—through collaborative cross-examination spanning methodology and theory.

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CONFLICT OF INTEREST STATEMENT

We do not have any conflicts of interest to declare.

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ENDNOTES

[1] Numerous studies report neural correlates of political cognition, but do not focus on the neural correlates of particular ideological orientations (e.g., Zamboni et al., 2009). We do not discuss such findings here due to our focus on the neural basis of ideology (see Jost et al., 2014 for discussions of political cognition studies).

[2] Due to a lack of socially conservative subjects, Pedersen et al. (2018) are limited to using economic conservatism as a predictor variable. Thus, the results of this study do not necessarily contradict theorizing and findings primarily focused on social conservatism (e.g., Hibbing et al., 2014; Tritt et al., 2013).

[3] Ahn et al. (2014) also report no differences in liberal and conservative responses to high-salience threatening images.

[4] It should also be noted that these relationships do not always hold and are sometimes reversed for economic conservatism (Costello et al., 2023; Malka et al., 2019).

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